

GO\_Biological\_Process\_2023 Top pathways by non-permutation

|  | Geneset                                   | stat        | num.genes | pval      | p.adj     | gene.vals                                                                 |
|--|-------------------------------------------|-------------|-----------|-----------|-----------|---------------------------------------------------------------------------|
|  | Monocarboxylic Acid Transport (GO:001571) | 0.17370502  | 61        | 2.770e-06 | 1.492e-02 | SLC16A5:36 SLC01B1:38 SLC5A12:96 SLC01A2:128 SLC22A13:218 SLC6A12:240     |
|  | Regulation Of Cell Population Proliferat  | -0.04988025 | 627       | 2.571e-05 | 6.922e-02 | AVPR2:30 SMARCC1:15 MAPK15:34 ERBB4:40 HAS2:93 TSPAN13:97                 |
|  | T Cell Differentiation In Thymus (GO:003  | -0.31699960 | 13        | 7.591e-05 | 6.934e-02 | PTPRC:30 RPK3:631 ZFP36L2:64 ZFP36L1:805 SHH:1287 LIG4:1446               |
|  | Bile Acid And Bile Salt Transport (GO:00  | 0.28702848  | 16        | 7.059e-05 | 6.934e-02 | SLCO1B1:38 SLC01A2:128 SLC27A5:257 SLC10A2:329 SLC51A:763 SLC10A3:946     |
|  | Cytoplasmic Translation (GO:0002181)      | 0.17043577  | 45        | 7.726e-05 | 6.934e-02 | UBA52:310 FTSJ1:319 RPS27A:489 RPS18:509 RPL34:537 RPS15A:712             |
|  | Organic Hydroxy Compound Transport (GO:0  | 0.19417580  | 35        | 7.089e-05 | 6.934e-02 | SLCO1B1:38 SLC01A2:128 ABCG8:232 SLC27A5:257 SLC10A2:329 RALBP1:455       |
|  | Positive Regulation Of DNA-templated Tra  | -0.03581660 | 1024      | 1.496e-04 | 1.151e-01 | E2F8:6 SMARCC1:15 PPARGC1A:21 ERBB4:40 EP300:46 HMGNI:52                  |
|  | Positive Regulation Of Cell Population P  | -0.05615917 | 383       | 1.811e-04 | 1.219e-01 | IL5RA:3 SMARCC1:15 PTPRC:30 MAPK15:34 ERBB4:40 HAS2:93                    |
|  | piRNA Processing (GO:0034587)             | 0.25739148  | 17        | 2.939e-04 | 1.275e-01 | PWIL1:33 GPAT2:150 MOV10L1:170 ANKRD34C:384 ASZ1:839 DD4:883              |
|  | Positive Regulation Of Transcription By   | -0.03938354 | 767       | 2.605e-04 | 1.275e-01 | E2F8:6 PPARGC1A:21 EP300:46 GLIS3:54 NCX2:158 MMP12:61                    |
|  | Positive Regulation Of Transcription Reg  | -0.06465646 | 16        | 2.466e-04 | 1.275e-01 | EP300:46 DDRG14:240 RB1:433 H1-0663 NEUROD1:844 PAX6:941                  |
|  | Positive Regulation Of Cellular Process   | -0.04859186 | 469       | 3.555e-04 | 1.497e-01 | SMARCC1:15 MAPK15:34 ERBB4:40 CITED2:68 CCR7:75 HAS2:93                   |
|  | Skin Development (GO:0043588)             | 0.14311359  | 52        | 3.613e-04 | 1.497e-01 | ADAM9:8 ACER1:117 OPN3:534 ITGB4:587 ST14:849 EVPL:1107                   |
|  | U4 snRNA 3'-End Processing (GO:0034475)   | -0.37094163 | 7         | 6.774e-04 | 1.967e-01 | EXOSC5:532 EXOSC7:1060 EXOSC8:1189 EXOSC9:1229 EXOSC3:1334 EXOSC6:1709    |
|  | Cellular Response To Growth Factor Stimu  | -0.08431641 | 135       | 7.413e-04 | 1.967e-01 | SEMA6A:90 HAS2:93 NTRK1:102 ANKRD1:248 COR01B:256 SPRY2:346               |
|  | Detection Of Chemical Stimulus Involved   | -0.14597299 | 47        | 5.411e-04 | 1.967e-01 | OR2T1:41 OR10G3:291 OR10K2:319 OR10T2:466 OR2S2:679 OR10G6:722            |
|  | Detection Of Chemical Stimulus Involved   | -0.06737207 | 46        | 5.991e-04 | 1.967e-01 | OR2T1:41 OR10G3:291 OR10K2:319 OR10T2:466 OR2S2:679 OR10G6:722            |
|  | Detection Of Chemical Stimulus Involved   | 0.20505737  | 22        | 8.732e-04 | 1.967e-01 | GNAT2:588 PKD1L3:619 TAS2R41:657 TAS2R10:801 TAS2R60:897 GNAT3:933        |
|  | Hepatocyte Differentiation (GO:0070365)   | -0.43463817 | 5         | 7.631e-04 | 1.967e-01 | E2F8:6 E2F7:646 PCK2:1139 PROX1:1214 PCK1:1803 NA                         |
|  | Keratinocyte Differentiation (GO:0030216  | 0.18507847  | 27        | 8.766e-04 | 1.967e-01 | ADAM9:8 OPN3:534 ST14:849 EVPL:1107 TGM1:1988 EXPH5:2540                  |
|  | Negative Regulation Of Blood Pressure (G  | 0.26716909  | 13        | 8.532e-04 | 1.967e-01 | DRD2:1074 DRD3:1214 PRCP:1346 GPR37L1:2023 NOS1:2624 RNL5:2852            |
|  | Nuclear-Transcribed mRNA Catabolic Proce  | -0.37094163 | 7         | 6.774e-04 | 1.967e-01 | EXOSC5:532 EXOSC7:1060 EXOSC8:1189 EXOSC9:1229 EXOSC3:1334 EXOSC6:1709    |
|  | Regulation Of ERK1 And ERK2 Cascade (GO:  | -0.06942455 | 195       | 8.686e-04 | 1.967e-01 | NEK10:23 PTPRC:30 ERBB4:40 CCR7:75 CCR1:79 SEMA6A:90                      |
|  | Sensory Perception Of Bitter Taste (GO:00 | 0.22889422  | 19        | 5.539e-04 | 1.967e-01 | GNAT2:588 TAS2R41:657 TAS2R10:801 CALHM1:860 TAS2R60:897 GNAT3:933        |
|  | NADH Dehydrogenase Complex Assembly (GO:  | 0.13669925  | 48        | 1.061e-03 | 1.997e-01 | TMEM126A:101 NDUFS5:196 ECSIT:253 NDUFB8:325 NDUFS3:356 NDUFA9:440        |
|  | Acid Secretion (GO:0046717)               | 0.33417053  | 8         | 1.064e-03 | 1.997e-01 | SLC22A16:70 UGT1A6:604 SLC51A:763 DRD3:1214 SLC51B:2740 SLC26A6:3639      |
|  | Amino Acid Transmembrane Transport (GO:0  | -0.17401039 | 41        | 1.134e-03 | 1.997e-01 | SLC6A6:9 SLC22A4:668 SLC7A9:774 SLC38A4:1193 SLC7A10:1362 SLC36A2:1506    |
|  | Mitochondrial Respiratory Chain Comple    | 0.13669925  | 48        | 1.061e-03 | 1.997e-01 | TMEM126A:101 NDUFS5:196 ECSIT:253 NDUFB8:325 NDUFS3:356 NDUFA9:440        |
|  | Phospholipid Translocation (GO:0045332)   | 0.18083229  | 27        | 1.150e-03 | 1.997e-01 | ATP9B:332 ANO6:514 ABCA4:997 ABCB1:1466 ATP8B2:1903 ATL11A:1917           |
|  | Regulation Of Organelle Organization (GO  | -0.10394287 | 84        | 1.109e-03 | 1.997e-01 | TOM1:17 MAPK15:34 EP300:46 PINK1:50 RAP1GDS1:67 MAPK3:205                 |
|  | Steroid Hormone Biosynthetic Process (GO  | 0.23541953  | 16        | 1.051e-03 | 1.997e-01 | CYP11A1:5 CYP17A1:11 CYP21A2:77 HSD17B2:390 HSD17B6:846 HSD17B3:1140      |
|  | Endoplasmic Reticulum Tubular Network O   | -0.24848175 | 14        | 1.288e-03 | 2.109e-01 | TMEM33:378 RTN1:507 RETREG3:509 RTN4:513 ATL1:752 ZFYVE27:1185            |
|  | Positive Regulation Of ERK1 And ERK2 Cas  | -0.07870628 | 141       | 1.292e-03 | 2.109e-01 | PTPRC:30 ERBB4:40 CCR7:75 CCR1:79 NTRK1:102 MAPK3:205                     |
|  | Detection Of Chemical Stimulus Involved   | 0.21695600  | 18        | 1.443e-03 | 2.230e-01 | GNAT2:588 TAS2R41:657 TAS2R10:801 TAS2R60:897 GNAT3:933 TAS2R1:2267       |
|  | Inorganic Cation Import Across Plasma Me  | 0.09571436  | 93        | 1.450e-03 | 2.230e-01 | SCNN1G:168 KCNJ1:188 ATP2B4:403 SCNN1B:469 TRPM1:612 SCNN1A:681           |
|  | Natural Killer Cell Differentiation (GO:  | -0.40929694 | 5         | 1.527e-03 | 2.283e-01 | PTPRC:30 PIK3CD:624 RABL3:1717 NFIL3:1936 TOX:2378 NA                     |
|  | Amino-Acid Betaine Transport (GO:0015838  | 0.34206847  | 7         | 1.724e-03 | 2.336e-01 | SLC22A16:70 SLC6A12:240 SLC22A4:668 SLC6A20:1870 SLC7A6:2117 SLC22A5:3732 |
|  | Circulatory System Development (GO:00723  | -0.08553614 | 113       | 1.722e-03 | 2.336e-01 | ERBB4:40 CITED2:68 PDLIM1:198 POU6F1:238 DNAAF4:306 HSPG2:313             |
|  | Intracellular Monoatomic Ion Homeostasis  | 0.20670617  | 19        | 1.818e-03 | 2.336e-01 | CLDN16:1 RHCG:91 CNMNA:194 RHAG:313 SLC4A1:625 LETMD1:1363                |
|  | Neuron Differentiation (GO:0030182)       | -0.07573388 | 143       | 1.822e-03 | 2.336e-01 | ERBB4:40 WNT9B:92 BRSK1:107 VAX1:164 FZD8:201 SPINK5:219                  |

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

|  | Geneset                                  | stat        | num.genes | pval      | p.adj     | gene.vals                                                                  |
|--|------------------------------------------|-------------|-----------|-----------|-----------|----------------------------------------------------------------------------|
|  | HOUNKPE_HOUSEKEEPING_GENES               | -0.04368203 | 853       | 1.798e-05 | 3.798e-02 | SNRNP70:4 PPP2R5A:10 SMARCC1:15 GBPBP1:71 TAOK2:74 SCYL1:81                |
|  | MILI_PSEUDOPODIA_HAPTOTAXIS_UP           | -0.06138209 | 407       | 2.342e-05 | 3.798e-02 | ARHGAP5:44 TRAPP8:47 GBBP1:71 R10K3:80 SCP2:91 TUT7:142                    |
|  | REACTOME_LIVER_OF_SMALL_MOLECULES        | 0.05052541  | 624       | 1.887e-05 | 3.798e-02 | CLCNKA:3 SLC6A6:9 TRPM6:17 CLCNKB:21 ANO5:31 SLC01B1:38                    |
|  | HSIAO_LIVER_SPECIFIC_GENES               | 0.08937665  | 196       | 1.668e-05 | 3.798e-02 | AGXT:27 ATP13A3:40 ALDH1A1:127 SORD:165 PROC:171 CYP27A1:231               |
|  | REACTOME_SARS_COV_2_MODULATES_HOST_TRANS | 0.21602218  | 31        | 3.156e-05 | 4.094e-02 | SNRPD3:375 RPS27A:489 RPS18:509 SNRPD1:656 RPS15A:712 RPS23:735            |
|  | MIKKELSEN_MEF_HCP_WITH_H3_UNMETHYLATED   | 0.08645661  | 186       | 4.941e-05 | 4.579e-02 | TRPM6:17 PWIL1:33 TLE6:61 FAR2:66 NSD1:76 KCNV2:142                        |
|  | FERREIRA_EWINGS_SARCOMA_UNSTABLE_VS_STAB | 0.10098979  | 136       | 4.893e-05 | 4.579e-02 | FANCM:156 CHAF1A:341 SNRPD3:375 IKBKB:432 NDUFA9:440 PINX1:448             |
|  | KIM_RESPONSE_TO_TSA_AND_DECITABINE_UP    | 0.11211014  | 105       | 7.337e-05 | 5.949e-02 | IL1R2:14 KCNV2:142 GDF15:569 ILIRAP:594 STAG3:643 SMPD3:724                |
|  | BLALOCK_ALZHEIMERS_DISEASE_DN            | -0.03710541 | 996       | 9.002e-05 | 6.488e-02 | ARHGEF3:66 PINK1:50 MRPL20:59 DGUOK:66 RAPIGDS1:67 SCP2:91                 |
|  | REACTOME_PROCESSING_OF_DNA_DOUBLE_STRAND | 0.13414338  | 70        | 1.051e-04 | 6.821e-02 | RNF168:78 UBA52:310 CLSPN:136 H2BC21:461 RPS27A:489 BRCA1:501              |
|  | REACTOME_OLFACTORY_SIGNALING_PATHWAY     | -0.11257732 | 93        | 1.777e-04 | 8.302e-02 | OR2T1:41 OR10G3:291 OR10K2:319 OR10T2:466 OR52B6:567 OR5E2:657             |
|  | REACTOME_BUTYRATE_RESPONSE_FACTOR_1_BRF1 | -0.26893947 | 16        | 1.910e-04 | 8.302e-02 | XRN1:148 EXOSC5:532 ZFP36L1:805 OCPIA1:1017 EXOSC7:1060 EXOSC8:1189        |
|  | WP_DRUG_INDUCTION_OF_BILE_ACID_PATHWAY   | 0.30249750  | 13        | 1.591e-04 | 8.302e-02 | ABCB11:35 SLC01B1:38 CYP7A1:690 SLC51A:763 NR1H4:776 SLC10A1:1356          |
|  | KIM_ALL_DISORDERS_CALB1_CORR_UP          | -0.05221443 | 439       | 1.897e-04 | 8.302e-02 | SNX17:14 UBQLN2:28 PINK1:50 AMPH:161 WDR47:228 FBXO9:277                   |
|  | DUTERTRE ESTRADIOL_RESPONSE_24HR_UP      | 0.06342925  | 294       | 1.920e-04 | 8.302e-02 | IQGAP3:86 KCNK15:92 DNMT1:105 ESCO2:113 UHRF1:176 HAUS8:185                |
|  | REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND | 0.43316026  | 6         | 2.382e-04 | 9.182e-02 | CA4:162 RHAG:313 CA1:496 SLC4A1:625 AQP1:1539 CA2:2933                     |
|  | REACTOME_SPERM_MOTILITY_AND_TAXES        | 0.39856712  | 7         | 2.603e-04 | 9.182e-02 | KCNU1:81 CATSPERG:183 CATSPERB:210 CATSPERD:423 CATSPERA:622 CATSPER3:3252 |
|  | REACTOME_ION_CHANNEL_TRANSPORT           | 0.08190641  | 166       | 2.783e-04 | 9.182e-02 | CLCNKA:3 TRPM6:17 CLCNKB:21 ANO5:31 ASIC5:89 TRPM5:109                     |
|  | WP_CYTOPLASMIC_RIBOSOMAL_PROTEINS        | 0.15646982  | 45        | 2.831e-04 | 9.182e-02 | UBA52:310 RPS27A:489 RPS18:509 RPL34:537 RPS15A:712 RPS23:735              |
|  | KEGG_STEROID_HORMONE_BIOSYNTHESIS        | 0.20024192  | 28        | 2.455e-04 | 9.182e-02 | CYP11A1:5 CYP17A1:11 SULT1E1:44 CYP1A1:87 COMT:122 CYP21A2:277             |
|  | WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE | 0.15218101  | 47        | 3.084e-04 | 9.525e-02 | NDUFS5:196 ECSIT:253 NDUFB8:325 NDUFS3:356 NDUFB5:502 NDUFB4:578           |
|  | REACTOME_STIMULI_SENSING_CHANNELS        | 0.10156946  | 102       | 3.987e-04 | 9.578e-02 | CLCNKA:3 TRPM6:17 CLCNKB:21 ANO5:31 ASIC5:89 TRPM5:109                     |
|  | REACTOME_TRISTETRAPROLIN_TTP_ZFP36_BINDS | -0.26463044 | 15        | 3.874e-04 | 9.578e-02 | XRN1:148 EXOSC5:532 DCP1A:1017 EXOSC7:1060 EXOSC8:1189 EXOSC9:1229         |
|  | REACTOME_PIWI_INTERACTING_RNA_PIRNA_BIOG | 0.24246914  | 18        | 3.691e-04 | 9.578e-02 | PWIL1:33 MOV10L1:170 PWIL2:239 POLR2J:417 POLR2I:560 ASZ1:839              |
|  | BRUINS_UVC_RESPONSE_VIA_TP53_GROUP_B     | -0.04474204 | 484       | 3.798e-04 | 9.578e-02 | IL5RA:3 TRIM3:101 CCDC127:152 CEP85L:196 COL4A1:209 QSOX1:212              |
|  | WEBER_METHYLATED_HCP_IN_FIBROBLAST_DN    | 0.17580017  | 34        | 3.900e-04 | 9.578e-02 | LPIN1:125 MOV10L1:170 DKKL1:323 BOLL:942 SYCP2:1417 TRPM4:1455             |
|  | WINTER_HYPOXIA_METAGENES                 | -0.07127746 | 210       | 3.820e-04 | 9.578e-02 | PLOD2:37 CTNNA4P:544 CITED2:68 R10K3:80 MAP4:150 TP1:171                   |
|  | FLORIO_NEOCORTEX_BASAL_RADIAL_GLIA_DN    | 0.07749059  | 172       | 4.659e-04 | 9.636e-02 | ANO5:31 SMC4:75 IQGAP3:86 ESCO2:113 MIS18BP1:119 KIF18B:135                |
|  | MEISSNER_NPC_HCP_WITH_H3_UNMETHYLATED    | 0.04715317  | 478       | 4.442e-04 | 9.636e-02 | TRPM6:17 ANO5:31 FAR2:66 RHCG:91 TPMRSS2:110 NAA1:157                      |
|  | REACTOME_REPRODUCTION                    | 0.10473207  | 95        | 4.242e-04 | 9.636e-02 | ADAM2:20 KCNU1:81 SYNE1:100 B4GALT1:129 CATSPERG:183 CATSPERB:210          |
|  | REACTOME_RECYCLING_OF_BILE_ACIDS_AND_SAL | 0.25190572  | 16        | 4.858e-04 | 9.636e-02 | ABCB11:35 SLC01B1:38 SLC01A2:128 SLC27A5:257 ALB:299 SLC10A2:329           |
|  | REACTOME_AMINO_ACID_TRANSPORT_ACROSS_THE | 0.18151895  | 31        | 4.702e-04 | 9.636e-02 | SLC6A6:9 SLC6A12:240 SLC43A1:737 SLC7A9:774 SLC6A19:1179 SLC38A4:1193      |
|  | BIOCARTA_H2AFISY_PATHWAY                 | -0.21976669 | 21        | 4.902e-04 | 9.636e-02 | COL4A1:209 F9:443 COL4A2:607 F2:777 F2R:802 FR:1176                        |
|  | CHANGOLKAR_H2AFY_TARGETS_UP              | 0.14888645  | 45        | 5.516e-04 | 1.022e-01 | SCD:189 CA1:496 SERPINAT:548 RORA:586 GLO1:618 ENPP3:644                   |
|  | REACTOME_PEPTIDE_LIGAND_BINDING_RECEPTOR | -0.07822108 | 165       | 5.385e-04 | 1.022e-01 | ACKR2:2 AVPR2:7 HEBP1:60 CCR7:75 CCR1:79 NPW:151                           |
|  | REACTOME_GENE_SILENCING_BY_RNA           | 0.08213380  | 85        | 5.691e-04 | 1.026e-01 | PWIL1:33 MOV10L1:170 PWIL2:239 POLR2J:417 H2BC21:461 POLR2I:560            |
|  | REACTOME_CARGO_CONCENTRATION_IN_THE_ER   | -0.19060542 | 27        | 6.089e-04 | 1.068e-01 | TMED10:220 SEC24C:708 LMAN2:859 FR:1176 LMAN2L:1221 STX5:1959              |
|  | KEGG_TASTE_TRANSDUCTION                  | 0.16631591  | 35        | 6.634e-04 | 1.133e-01 | TRPM5:109 SCNN1G:168 SCNN1B:469 TAS1R1:638 TAS2R41:657 TAS2R10:801         |
|  | KEGG_VALINE_LEUCINE_AND_ISOLEUCINE_DEGRA | 0.14940108  | 43        | 7.024e-04 | 1.168e-01 | EHHADH:22 BCAT2:47 ACAD8:314 ACADM:590 IL4I1:809 HIBCX:952                 |
|  | REACTOME_VITAMIN_D_CALCIFEROL_METABOLISM | 0.30632133  | 10        | 7.956e-04 | 1.259e-01 | CUBN:201 UBE2I:507 LGMN:1629 CYP2R1:2123 VDR:2625 P1A5A:2982               |

DisGeNET Top pathways by non-permutation

|  | Geneset                                  | stat        | num.genes | pval      | p.adj     | gene.vals                                                           |
|--|------------------------------------------|-------------|-----------|-----------|-----------|---------------------------------------------------------------------|
|  | Renal salt wasting                       | 0.36698975  | 15        | 8.637e-07 | 8.469e-03 | CLCNKA:3 CYP11A1:5 CLCNKB:21 SCNN1G:168 KCNJ1:188 CYP21A2:277       |
|  | Renal tubular acidosis                   | 0.20636568  | 35        | 2.412e-05 | 1.183e-01 | CLDN16:1 TRPM6:17 CLCNKB:21 EHHAHD:22 KCNJ1:188 MRGPRF:268          |
|  | Hyperkalemia                             | 0.27341192  | 17        | 9.233e-05 | 2.991e-01 | CYP11A1:5 CYP17A1:11 LPIN1:125 SCNN1G:168 SCNN1B:469 WNK4:831       |
|  | Nephrocalcinosis                         | 0.15414757  | 52        | 1.220e-04 | 2.991e-01 | CLDN16:1 TRPM6:17 CLCNKB:21 AGXT:27 NSD1:76 SLC26A3:172             |
|  | Hypochloremia (disorder)                 | 0.43763046  | 6         | 2.054e-04 | 3.356e-01 | CLCNKA:3 CLCNKB:21 SLC26A3:172 KCNJ1:188 BSND:1259 SLC12A1:3855     |
|  | Serum chloride level decreased (finding) | 0.43763046  | 6         | 2.054e-04 | 3.356e-01 | CLCNKA:3 CLCNKB:21 SLC26A3:172 KCNJ1:188 BSND:1259 SLC12A1:3855     |
|  | Hyperactive renin-angiotensin system     | 0.36970772  | 8         | 2.934e-04 | 3.597e-01 | CLCNKB:21 SCNN1G:168 SLC26A3:172 KCNJ1:188 SCNN1B:469 SCNN1A:2227   |
|  | Increased plasma renin activity          | 0.36970772  | 8         | 2.934e-04 | 3.597e-01 | CLCNKB:21 SCNN1G:168 SLC26A3:172 KCNJ1:188 SCNN1B:469 SCNN1A:2227   |
|  | Cholestyrolithiasis                      | 0.13037476  | 61        | 4.345e-04 | 4.734e-01 | CYP17A1:11 ABCB11:35 SLC01B1:38 CYP1A1:87 SLC01A2:128 ABCG8:232     |
|  | Atrophy, Disuse                          | -0.36593425 | 7         | 8.002e-04 | 5.276e-01 | PPARGC1A:21 HSPA4:134 PITX1:1442 PPARGC1B:1907 LIF:1964 FBXO32:3063 |
|  | Abnormality of macular pigmentation      | 0.20394742  | 21        | 1.218e-03 | 5.276e-01 | PROM1:41 GPR179:281 TRPM1:612 PDE6B:973 ABCA4:997 GRM6:1229         |
|  | Abnormality of the urinary system        | 0.20366478  | 21        | 1.237e-03 | 5.276e-01 | GLB1:305 HGD:566 RGRIP1L:703 COL3A1:1451 RPS27:1555 RPS19:1607      |
|  | Acute kidney injury                      | 0.11240310  | 72        | 9.881e-04 | 5.276e-01 | A2M:77 HEXB:124 LPIN1:125 ALB:299 AFM:415 AMBP:678                  |
|  | Benign recurrent intrahepatic cholestasi | 0.38501572  | 6         | 1.091e-03 | 5.276e-01 | ABCB11:35 GTGLC3:198 GTGLC1:851 GGT2:1819 GGT1:2029 ATP8B1:5230     |
|  | Complicated pneumonococci                | 0.32293117  | 9         | 7.948e-04 | 5.276e-01 | SLC06A1:132 GSTK1:544 IL6:778 VEGFA:1734 ICAM1:1981 IL1A:2904       |
|  | Decreased glomerular filtration rate     | 0.35590228  | 7         | 1.111e-03 | 5.276e-01 | CLCNKA:3 CLCNKB:21 AGXT:27 BSND:1259 G6PC:2694 ITGA3:2798           |
|  | Generalized glycogen storage disease of  | 0.22966542  | 17        | 1.046e-03 | 5.276e-01 | SI:83 UGT1A6:604 GLA:865 ARSB:1106 DES:2681 G6PC:2694               |
|  | Glycinuria                               | 0.27374205  | 13        | 6.331e-04 | 5.276e-01 | ACADM:59 SLC6A19:1179 GCSH:1187 PCCB:3137 SLC36A2:1506 SLC6A20:1870 |
|  | Hyperchloremiduria                       | 0.42751711  | 5         | 9.306e-04 | 5.276e-01 | CLCNKA:3 CLCNKB:21 KCNJ1:188 BSND:1259 SLC12A1:3855 NA              |
|  | HYPERGLYCINURIA (disorder)               | 0.42540271  | 15        | 6.596e-04 | 5.276e-01 | ACADM:59 SLC6A19:1179 GCSH:1187 PCCB:3137 SLC6A2:1506 SLC6A20:1870  |
|  | Increased urinary potassium              | 0.42751711  | 5         | 9.306e-04 | 5.276e-01 | CLCNKA:3 CLCNKB:21 KCNJ1:188 BSND:1259 SLC12A1:3855 NA              |
|  | Kidney Calculi                           | 0.13523541  | 53        | 6.670e-04 | 5.276e-01 | CLDN16:1 AGXT:27 SLC01B1:38 SLCNA1:625 AMBP:678 IL6:778             |
|  | Macular pigmentary changes               | 0.20394742  | 21        | 1.218e-03 | 5.276e-01 | PROM1:41 GPR179:281 TRPM1:612 PDE6B:973 ABCA4:997 GRM6:1229         |
|  | Disorder of the genitourinary system     | 0.34870932  | 7         | 1.399e-03 | 5.715e-01 | PLEC:88 ITGB5:587 SFRP1:1132 ITGA6:1286 FSHR:1338 MED12:4886        |
|  | Turner Syndrome, Male                    | -0.14565442 | 38        | 1.901e-03 | 7.170e-01 | XHC:125 MAPK3:205 GH1:464 GPNMB:471 ACP1:649 MPN1:483               |
|  | Cholestasis                              | 0.05770888  | 248       | 1.838e-03 | 7.170e-01 | CYP17A1:11 ABCB11:35 SLC01B1:38 SLC23A1:39 AFP:104 GTGLC3:198       |
|  | Skin Manifestations                      | 0.26836821  | 11        | 2.058e-03 | 7.473e-01 | CYP1A1:87 PLEC:88 ACAD:314 NTSE:1686 GTDL:1716 IL13:1724            |
|  | Metabolic Bone Disorder                  | 0.16480128  | 29        | 2.137e-03 | 7.485e-01 | KDM4B:43 CYP27A1:231 GLB1:305 IL6:778 PTH:840 PHEX:1181             |
|  | Progressive intrahepatic cholestasis (di | 0.18665729  | 22        | 2.420e-03 | 7.910e-01 | ABCB11:35 GTGLC3:198 SLC01A2:329 NR1H4:776 H1L27B6:846 GTGLC1:851   |
|  | Rare Diseases                            | 0.23436139  | 14        | 2.399e-03 | 7.910e-01 | AGXT:27 ARG1:361 HGD:566 DNMT3:866 ABCB6:1019 CPS1:2488             |
|  | Acute myeloid leukaemia, minimal differ  | 0.19240684  | 18        | 2.841e-02 | 8.056e-01 | ELL:53 ZMYND11:1032 CD34:1406 ABCB1:1466 CD19:2599 NCPL1:2619       |
|  | Alport Syndrome, Autosomal Recessive     | -0.25368175 | 5         | 9.474e-02 | 8.056e-01 | COL4A4:2130 COL4A3:2389 MPV17:3406 COL4A5:4338 MYH9:6184 NA         |
|  | Amotyrophy, monomelic                    | 0.26657205  | 5         | 3.899e-02 | 8.056e-01 | COLM1:2667 SOD1:2199 SMDN2:2406 CPE126:532 CLPLANE1:5702 NA         |
|  | Anemia, Diamond-Blackfan                 | 0.09184676  | 45        | 3.315e-02 | 8.056e-01 | THPO:83A LPIN2:941 CD38:1127 RPL131:321 CD34:1406 COL3A1:1451       |
|  | Anemia, hereditary spherocytic hemolytic | 0.35518450  | 5         | 5.949e-03 | 8.056e-01 | EPB42:141 ANK1:429 SPTA1:547 SLC4A1:625 SPTB:9048 NA                |
|  | Anemia, Sickle Cell                      | 0.04416719  | 170       | 4.764e-02 | 8.056e-01 | ICAM4:37 COMT:122 RHD:147 CRYZ1:54 SCD:189 MUC4:202                 |
|  | ARTHROGRYPOSIS MULTIPLEX CONGENITA, DIST | 0.17989876  | 9         | 6.167e-02 | 8.056e-01 | WNK4:831 FBN2:1163 SLC12A3:1485 WNK1:2390 PIEZ2:2451 REN:5806       |
|  | Asthma, Occupational                     | 0.14966424  | 17        | 3.268e-02 | 8.056e-01 | TENM3:273 TACR2:666 PTGS2:1953 PKRCA:2988 TNF:3235 NAT2:3822        |
|  | Ataxias, Hereditary                      | 0.04865996  | 50        | 4.023e-02 | 8.056e-01 | RNF216:69 SYNE1:100 DNMT1:105 HEXB:124 CYP27A1:231 SNAP25:391       |
|  | Brain Injuries, Focal                    | 0.08694959  | 156       | 2.446e-02 | 8.056e-01 | ILB:299 IL17B:1132 IL13RA2:1082 IL1R1:1194 GABRB3:1256 RIF1:1311    |